

SIH 2026 · PS 26039 · Team FAB6 — Component List - ID 142638

Ref	Category	Component	Spec / variant	Qty	Unit ₹	Line ₹
A. Gas & environment						
01	A. Gas & environment	MQ-2 gas sensor module	Combustible gas / smoke, analog	1	₹150	₹150
02	A. Gas & environment	MQ-4 gas sensor module	Methane (CH ₄) specific, analog	1	₹180	₹180
03	A. Gas & environment	MQ-7 gas sensor module	Carbon monoxide, analog	1	₹150	₹150
04	A. Gas & environment	DHT22 / AM2302	Temp + humidity, 1-wire	1	₹250	₹250
B. Motion, range, thermal						
05	B. Motion, range, thermal	MPU6050 (GY-521)	6-axis IMU, I2C 0x68	1	₹110	₹110
06	B. Motion, range, thermal	HC-SR04 ultrasonic	Trigger/echo, 5 V	1	₹80	₹80
07	B. Motion, range, thermal	MLX90614 (GY-906)	IR thermometer, I2C 0x5A	1	₹700	₹700
08	B. Motion, range, thermal	SG90 micro servo	9 g, 50 Hz PWM	1	₹99	₹99
C. Compute & comms						
09	C. Compute & comms	ESP32 DevKit V1	WROOM-32, 30-pin	2	₹435	₹870
10	C. Compute & comms	LoRa Ra-02 (SX1278)	433 MHz SPI module	2	₹280	₹560
11	C. Compute & comms	433 MHz antenna	SMA whip or 17.3 cm spring	2	₹90	₹180
12	C. Compute & comms	Micro-USB data cable	Data-capable, not charge-only	2	₹80	₹160
D. Camera						
13	D. Camera	ESP32-CAM with CH340	OV2640 2 MP + onboard USB programmer	1	₹599	₹599
14	D. Camera	Electrolytic cap 220 uF	Local decoupling at the ESP32-CAM	1	₹10	₹10
E. Power						
15	E. Power	UBEC 5 V / 3 A switching	Input 2-6S	1	₹269	₹269
16	E. Power	AMS1117-3.3 module	5 V to 3.3 V LDO breakout	2	₹45	₹90
17	E. Power	XT60 male pigtail	14 AWG battery tap	1	₹60	₹60
18	E. Power	Polyfuse 2 A (MF-R200)	Resettable PTC, pack of 5	1	₹50	₹50
19	E. Power	IRF4905 P-MOSFET	TO-220, reverse-polarity protection	1	₹45	₹45
20	E. Power	TVS diode SMBJ15A	Transient clamp	1	₹40	₹40

F. Signal conditioning						
21	F. Signal conditioning	MCP6004 quad op-amp	DIP-14, rail-to-rail	1	₹70	₹70
22	F. Signal conditioning	DIP-14 IC socket	For the MCP6004	2	₹10	₹20
23	F. Signal conditioning	IRLZ44N N-MOSFET	Logic-level, TO-220	2	₹60	₹120
G. Passives						
24	G. Passives	Electrolytic cap set	MUST include 1000uF, 470uF/25V, 220uF, 100uF, 10uF	1	₹200	₹200
25	G. Passives	Resistor kit, 1% metal film	Needs 100k, 10k, 4.7k, 3.3k, 2.2k, 1k, 100R	1	₹250	₹250
26	G. Passives	Ceramic cap 100 nF	0.1 uF, pack of 50	1	₹60	₹60
H. Mechanical						
27	H. Mechanical	Payload pod	Vented floor + side louvres	1	₹0	₹0
28	H. Mechanical	Collimator tube	6 mm ID x 20 mm, matte black inside	1	₹0	₹0
29	H. Mechanical	Extended landing gear	Taller legs to clear the pod	1	₹0	₹0
30	H. Mechanical	M3 nylon standoff + screw kit	Assorted lengths	1	₹150	₹150
31	H. Mechanical	Velcro battery straps	300 mm	2	₹40	₹80
32	H. Mechanical	Zip ties + foam tape	Assorted	1	₹100	₹100
I. Prototyping						
33	I. Prototyping	Perfboard / zero PCB	General purpose	2	₹50	₹100
34	I. Prototyping	Berg strip headers	Male + female, 40-pin	4	₹20	₹80
35	I. Prototyping	Silicone wire 28 + 22 AWG	Stranded, multi-colour	1	₹250	₹250
36	I. Prototyping	Breadboard 830-point	Solderless	1	₹150	₹150
37	I. Prototyping	Jumper wire set	M-M, M-F, F-F	1	₹250	₹250
38	I. Prototyping	Heat shrink assortment	Assorted diameters	1	₹100	₹100
39	I. Prototyping	Solder wire + flux	60/40, 0.8 mm	1	₹250	₹250
J. Spares						
40	J. Spares	Spare ESP32 DevKit V1	3rd board	1	₹435	₹435
K. Tools (must have access)						
41	K. Tools (must have access)	Digital multimeter	Continuity + DC volts	1	₹0	₹0
42	K. Tools (must have access)	Soldering iron + stand	Temperature controlled preferred	1	₹0	₹0
43	K. Tools (must have access)	LiPo balance charger + safe bag	3S capable	1	₹0	₹0
44	K. Tools (must have access)	Bench 5 V supply / USB power bank	For the 24-48 h MQ burn-in	1	₹0	₹0

CASH NEEDED NOW (status = BUY)
IF THE LAB HAS NOTHING (BUY + LAB)
LAB-STOCK VALUE (status = CHECK LAB)

₹7,317
₹7,317
₹0

Status	Got it?	Why it is on this list
BUY		Heater 33 Ω , 152 mA continuous on 5 V
BUY		Primary firedamp channel
BUY		Needs the IRLZ44N below to cycle its heater
BUY		Run at 3.3 V. Mount 60 mm+ from the MQ cluster
BUY		Vibration sensing. Mount RIGID — no damping
BUY		ECHO is 5 V — needs the 2.2k/3.3k divider
BUY		LONG LEAD ITEM — order first, often out of stock
BUY		Thermal scan mount. Needs the 1000 uF cap
BUY		QTY 2 — one flies, one is the ground station
BUY		QTY 2. Now your ONLY comms link — tether deferred
BUY		NEVER power an SX1278 without one
BUY		Everyone has these. Check before buying
BUY		Replaces the ₹5,471 FPV chain. CH340 version = no FTDI needed
BUY		Wi-Fi TX pulses otherwise disturb the 5 V rail
BUY		Single rail now — no VTX means no second BEC
BUY		3.3 V rail for LoRa + sensors. 1 spare
BUY		Payload taps the pack separately from the PDB
BUY		Payload fault cannot reach the flight controller
BUY		One miswire otherwise kills the whole payload
BUY		Clamps ESC spikes on the payload branch

BUY		Buffers all 3 gas dividers — one chip does all of them
BUY		Never solder the IC straight to perfboard
BUY		MQ-7 heater cycling. Without it MQ-7 cannot read CO. 1 spare
BUY		Do not skip. No 1000 uF = the servo resets your ESP32
BUY		Standard lab stock. Verify these values are present
BUY		One at every active device VCC pin
BUY		Filament only (~₹250 if your dept charges)
BUY		Or a blackened brass tube / pen barrel, ~₹80
BUY		Printed legs are fine at 1.4 kg. Buy F450 tall gear (₹250) if unsure
BUY		Common lab stock
BUY		Battery slides fore/aft to trim the CG
BUY		Cable management and board mounting
BUY		Signal conditioning board. Gets consumed
BUY		Socket the modules — do not solder them down
BUY		Lab wire is usually solid core — cracks under vibration
BUY		For stage-by-stage bring-up
BUY		Standard lab stock
BUY		Every flying joint
BUY		Standard lab stock
BUY		The one spare worth buying. You WILL kill a board
BUY		MANDATORY — measure every divider before connecting it
BUY		
BUY		Safety critical
BUY		Burn-in starts DAYS before the demo